

KHATANGA, Russian Federation

WMO: 208910

Lat: **71.977N**Lon: **102.466E**Elev: **33**StdP: **100.93**Time Zone: **7.00 (E07)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-47.1	-44.9	-49.1	0.0	-47.0	-47.0	0.0	-44.7	14.2	-16.9	12.7	-18.5	2.3	50	0.970

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.2	24.3	16.6	21.9	15.3	19.4	14.0	17.5	23.2	15.9	21.1	14.3	18.7	4.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.0	10.7	20.2	13.3	9.6	18.3	11.9	8.7	17.1	49.6	23.3	44.8	21.2	40.4	18.8	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
11.8	9.8	8.3	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-49.1	28.6	2.7	2.4	-51.0	30.3	-52.5	31.7	-54.0	33.0	-55.9	34.7
			-49.1	19.6	2.7	1.5	-51.0	20.6	-52.5	21.5	-54.0	22.3	-55.9	23.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-11.6	-30.7	-32.0	-25.3	-15.7	-5.2	7.3	12.9	9.6	2.2	-11.2	-23.6	-29.2	(6)
(7)		DBStd	17.50	9.06	8.33	9.40	7.40	5.84	5.75	4.56	3.88	4.47	7.42	7.70	8.49	(7)
(8)		HDD10.0	8083	1260	1176	1093	771	472	119	21	56	236	656	1008	1216	(8)
(9)		HDD18.3	10945	1518	1409	1352	1021	731	335	179	271	483	914	1258	1474	(9)
(10)		CDD10.0	196	0	0	0	0	1	39	110	44	2	0	0	0	(10)
(11)		CDD18.3	15	0	0	0	0	0	4	10	1	0	0	0	0	(11)
(12)		CDH23.3	113	0	0	0	0	0	41	69	3	0	0	0	0	(12)
(13)		CDH26.7	21	0	0	0	0	0	10	11	0	0	0	0	0	(13)
(14)	Wind	WSAvg	4.3	4.3	4.0	4.2	4.4	4.6	4.7	4.4	4.3	4.4	4.1	3.9	3.9	(14)
(15)	Precipitation	PrecAvg	282	14	13	14	14	17	32	40	38	34	29	20	17	(15)
(16)		PrecMax	416	40	33	54	44	36	114	112	93	85	66	50	92	(16)
(17)		PrecMin	158	1	2	2	0	2	0	5	4	1	10	4	0	(17)
(18)		PrecStd	55	10	8	9	9	9	26	20	22	17	13	10	14	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-7.1	-6.1	-0.7	6.0	11.9	28.0	28.2	24.0	17.0	5.2	-2.7	-7.4	(19)
(20)			MCWB	-7.6	-6.5	-1.8	2.9	7.5	17.4	18.6	16.3	11.2	3.4	-3.3	-8.0	(20)
(21)		2%	DB	-10.9	-11.1	-3.8	3.0	7.8	23.4	25.2	21.2	13.0	2.3	-7.1	-10.1	(21)
(22)			MCWB	-11.3	-11.5	-4.6	1.0	4.2	15.7	17.3	14.7	9.1	1.4	-7.5	-10.5	(22)
(23)		5%	DB	-13.9	-15.2	-6.8	0.0	5.2	20.0	23.0	18.6	10.1	0.8	-9.7	-12.9	(23)
(24)			MCWB	-14.3	-15.6	-7.6	-1.4	2.5	13.7	16.1	13.8	7.6	0.3	-10.2	-13.3	(24)
(25)		10%	DB	-17.2	-19.3	-10.3	-3.9	2.9	16.2	20.7	16.1	8.0	-1.0	-12.5	-16.2	(25)
(26)			MCWB	-17.4	-19.5	-10.9	-5.0	1.1	11.4	15.1	12.4	6.2	-1.7	-13.0	-16.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.5	-6.6	-1.6	3.1	7.7	18.2	20.1	17.5	12.1	4.0	-3.4	-7.9	(27)
(28)			MCDWB	-7.4	-6.3	-0.6	5.6	11.2	26.1	26.1	21.5	16.1	4.8	-2.7	-7.6	(28)
(29)		2%	WB	-11.3	-11.6	-4.6	1.1	4.5	16.2	18.2	15.5	9.4	1.5	-7.6	-10.6	(29)
(30)			MCDWB	-11.0	-11.2	-3.8	2.8	7.1	22.8	23.4	19.4	12.1	2.0	-7.3	-10.2	(30)
(31)		5%	WB	-14.4	-15.6	-7.6	-1.4	2.7	14.1	16.9	14.1	7.8	0.2	-10.1	-13.4	(31)
(32)			MCDWB	-14.0	-15.3	-6.9	0.0	4.8	19.6	22.3	17.7	9.8	0.8	-9.7	-13.0	(32)
(33)		10%	WB	-17.6	-19.6	-10.9	-5.0	1.2	11.5	15.3	12.8	6.4	-1.8	-13.0	-16.6	(33)
(34)			MCDWB	-17.3	-19.4	-10.2	-3.9	2.6	16.0	20.2	16.2	7.8	-1.0	-12.6	-16.3	(34)
(35)	Mean Daily Temperature Range		MDBR	6.9	6.7	8.4	9.7	6.8	6.8	7.2	6.7	5.1	5.9	6.6	6.9	(35)
(36)		5% DB	MCDBR	9.4	9.4	8.1	10.5	8.4	11.7	10.8	11.0	9.0	5.5	8.9	9.6	(36)
(37)			MCWBR	9.3	9.3	7.7	9.1	5.9	6.4	5.7	6.1	5.9	4.9	8.8	9.4	(37)
(38)		5% WB	MCDWB	9.5	9.5	8.0	10.6	8.1	11.3	10.1	9.5	8.0	5.3	8.8	9.6	(38)
(39)			MCWBR	9.3	9.4	7.6	9.2	5.9	6.8	6.0	6.1	5.6	5.1	8.7	9.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	N/A	0.203	0.289	0.313	0.343	0.346	0.383	0.381	0.306	0.210	N/A	N/A	(40)	
(41)		taud	N/A	1.942	2.010	2.072	2.121	2.423	2.352	2.298	2.410	1.998	N/A	N/A	(41)	
(42)		Ebn at Noon	0	419	648	787	814	837	779	718	677	386	0	0	(42)	
(43)		Edh at Noon	0	33	80	108	119	93	97	89	56	28	0	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.00	0.22	1.67	4.18	5.88	5.50	4.90	2.98	1.38	0.40	0.00	0.00	(44)	
(45)		RadStd	0.00	0.02	0.14	0.21	0.48	0.49	0.43	0.39	0.16	0.06	0.00	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days			
			99% DB	99% DP	1% DB	1% WB	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
(46) Station Only		+1.14	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page